Mooncake

Trading More Storage for Less Computation – A KVCache-centric Architecture for Serving LLM Chatbot

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Moonshot AI

清華大學 Tsinghua University

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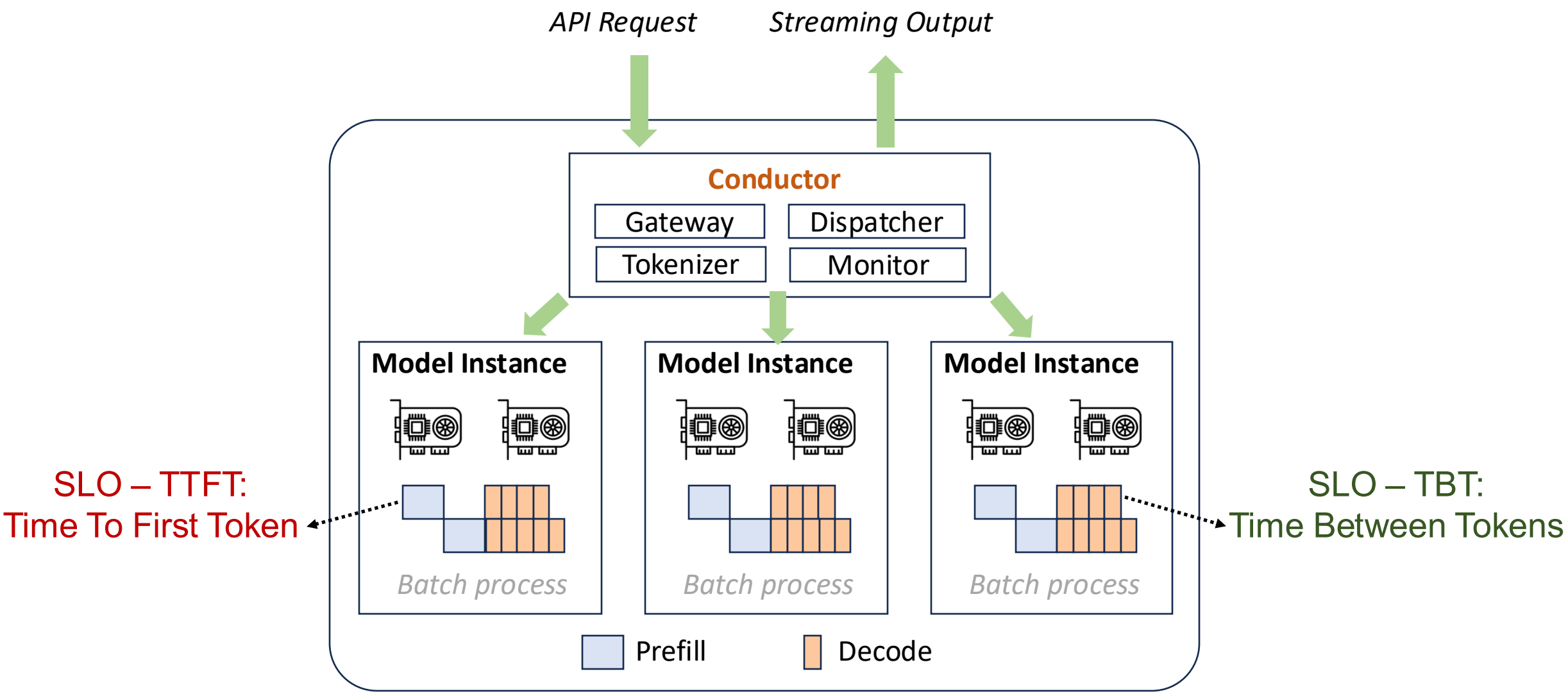
FAST'25 | Best Paper Award

Paper

GitHub

Online LLM Serving System



Online LLM Serving System

Kimi: a leading AI service provided by **Moonshot AI**

- Process over 100 billion tokens daily
- Real-time response
- Up to **1M tokens** context

More Data + Larger Model + Longer Context = ***Higher Intelligence***

Online LLM Serving System

More Data + Larger Model + Longer Context = ***Higher Intelligence***

BUT

Lack of GPU Supply

Higher Inference Cost

Longer Response Time

Online LLM Serving System

More Data + Larger Model + Longer Context = ***Higher Intelligence***

Reduce the cost for large-scale long-context LLM inference!

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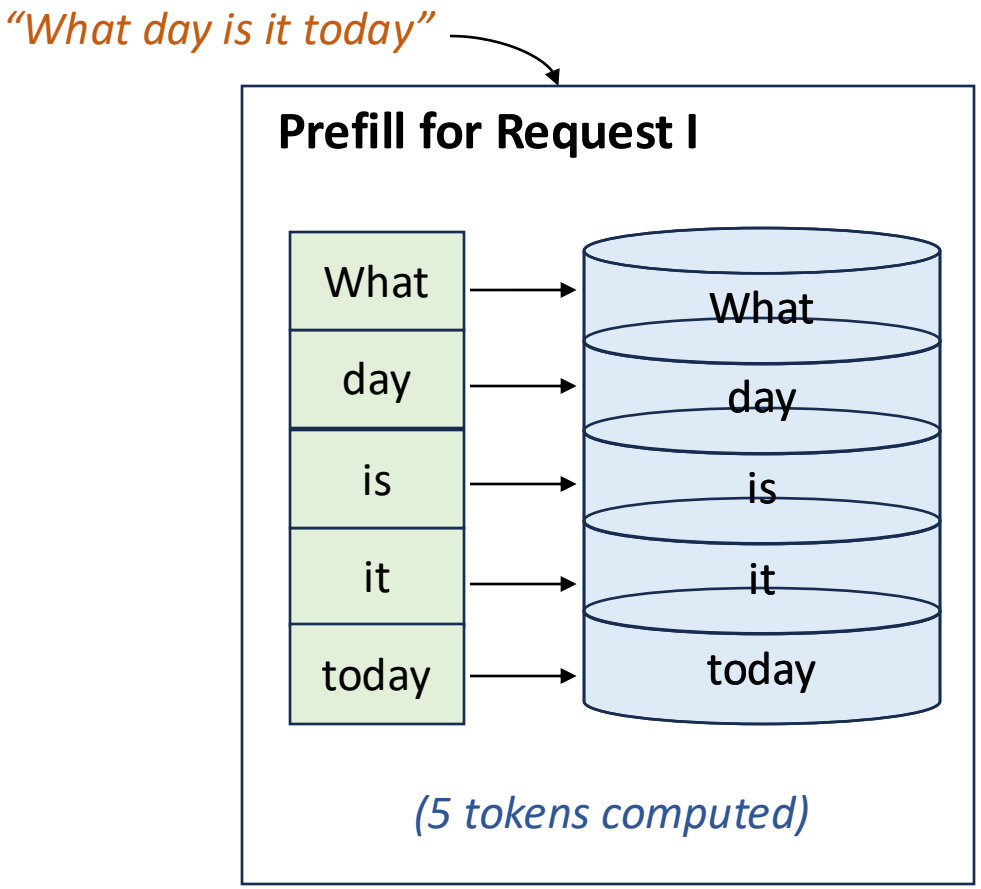
LLM Inference: Prefix Caching

- KVCache can be shared across requests with the same prefix, reducing computation during prefill



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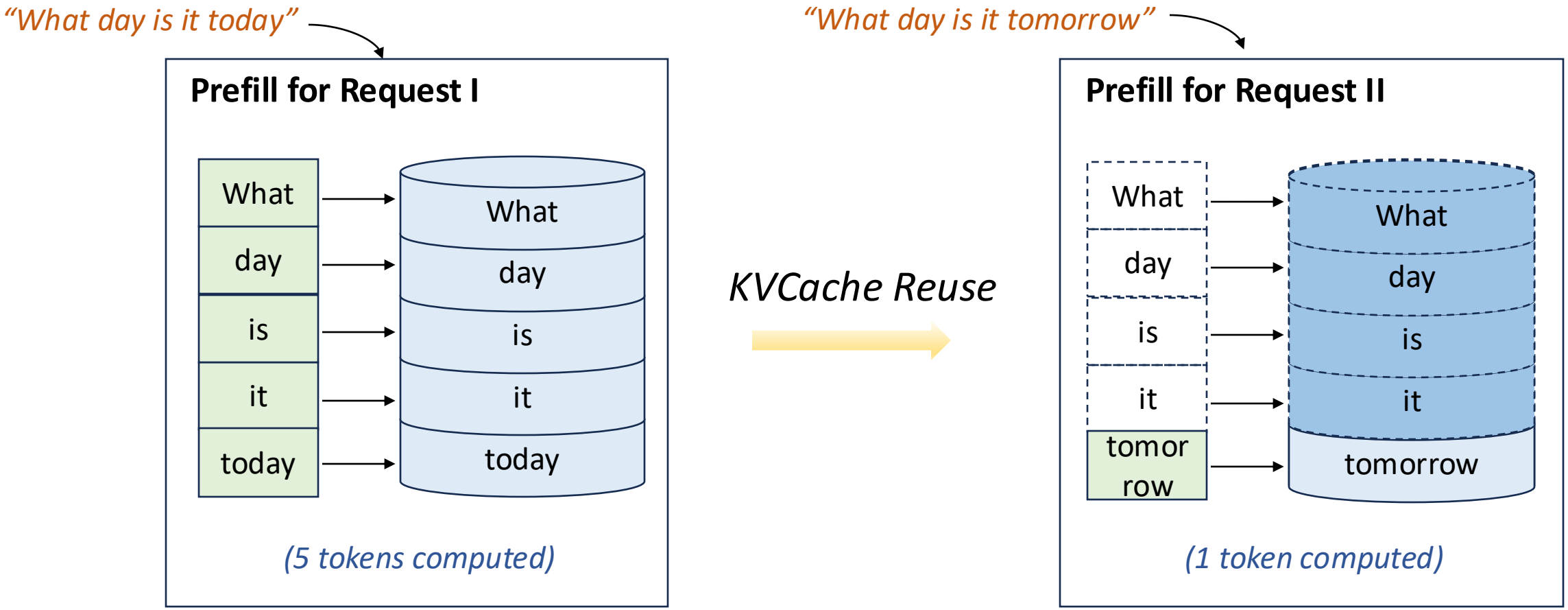
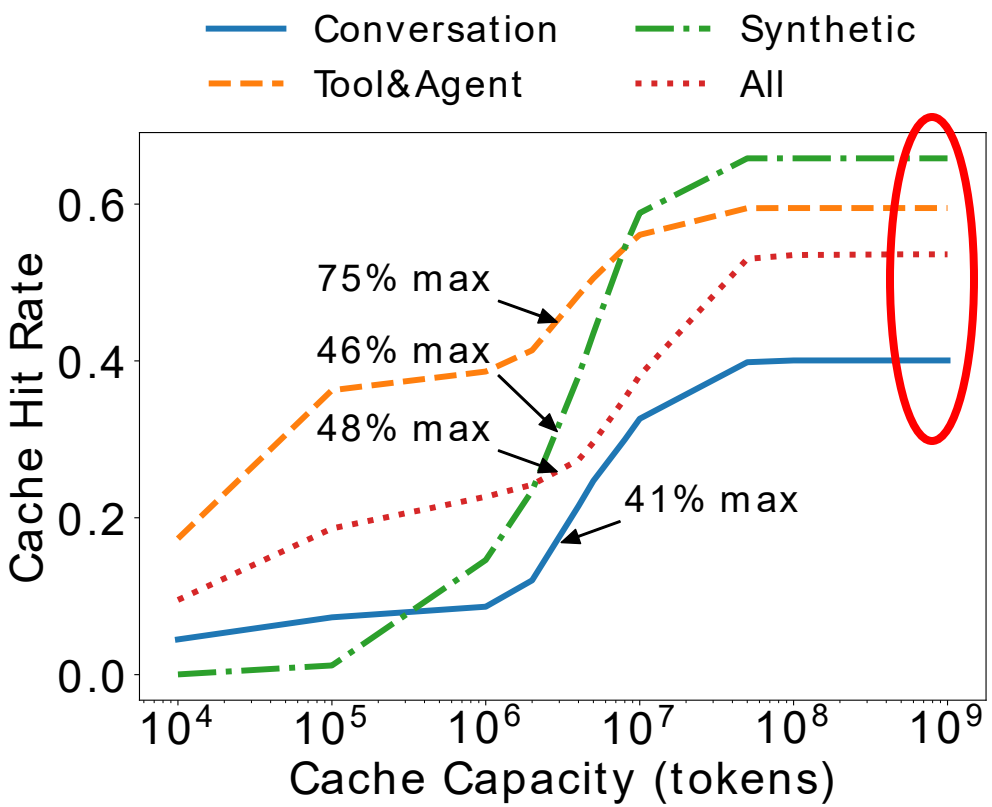


Trace Analysis: Is Prefix Caching a Simple Solution?



Traces in our paper
(open-sourced in https://github.com/kvcache-ai/Mooncake)

Conversation: collected from real-world online conversation requests

Tool&Agent: collected from real-world online requests that include tool use

Synthetic: synthesized from publicly available long context datasets

- Around **50%** of the tokens' KVCache in the real-world workloads can be reused

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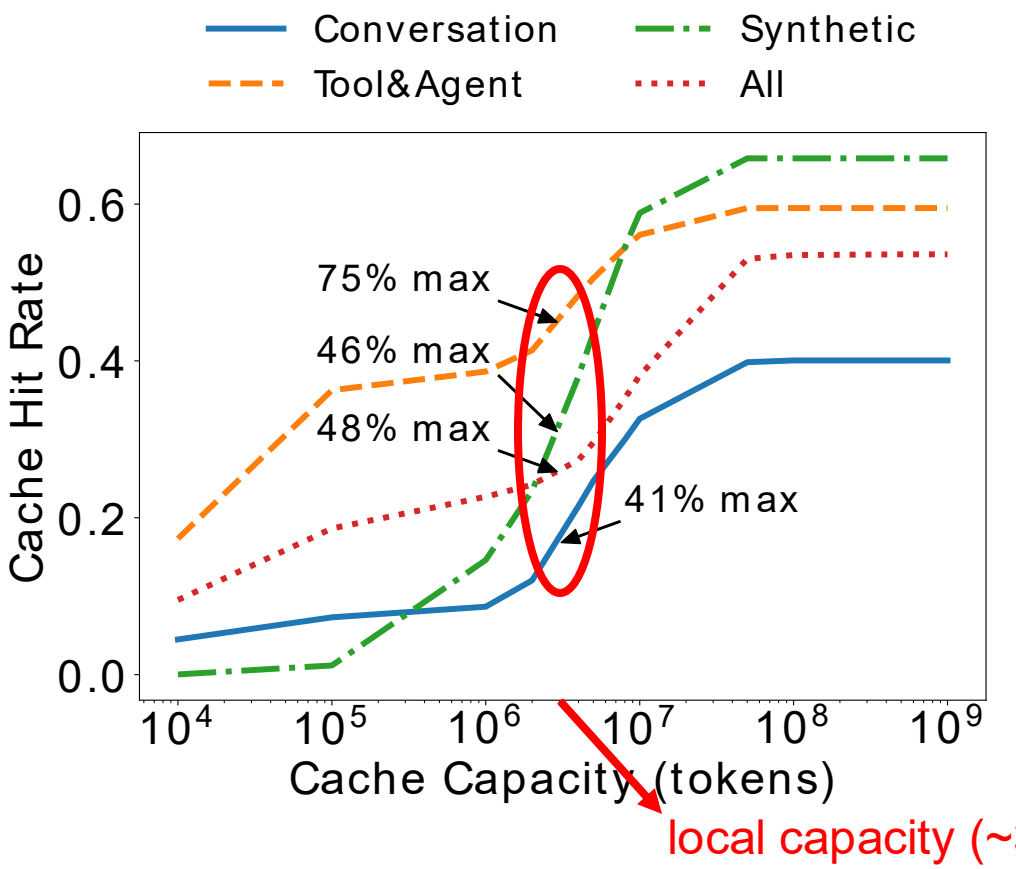

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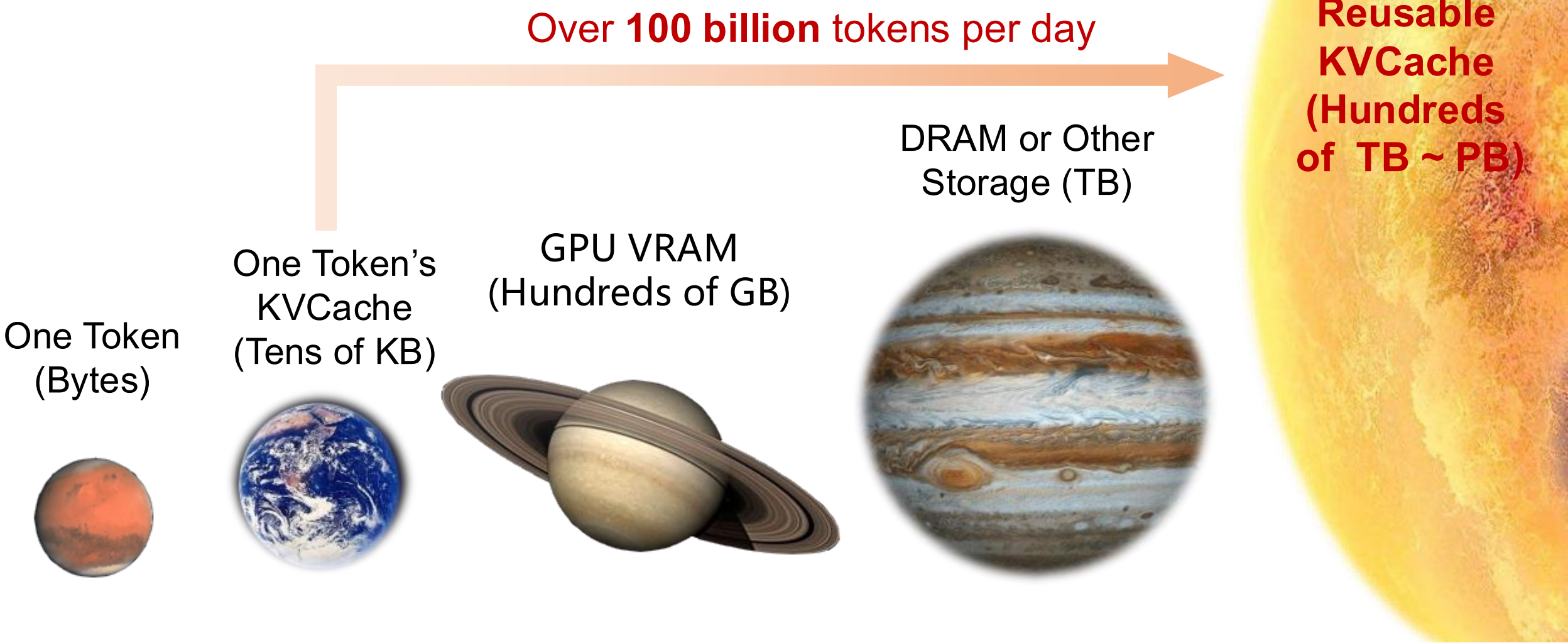
- Around **50%** of the tokens' KVCache in the real-world workloads can be reused
- **However**, the cache hit rate will significantly drop if only using the local cache

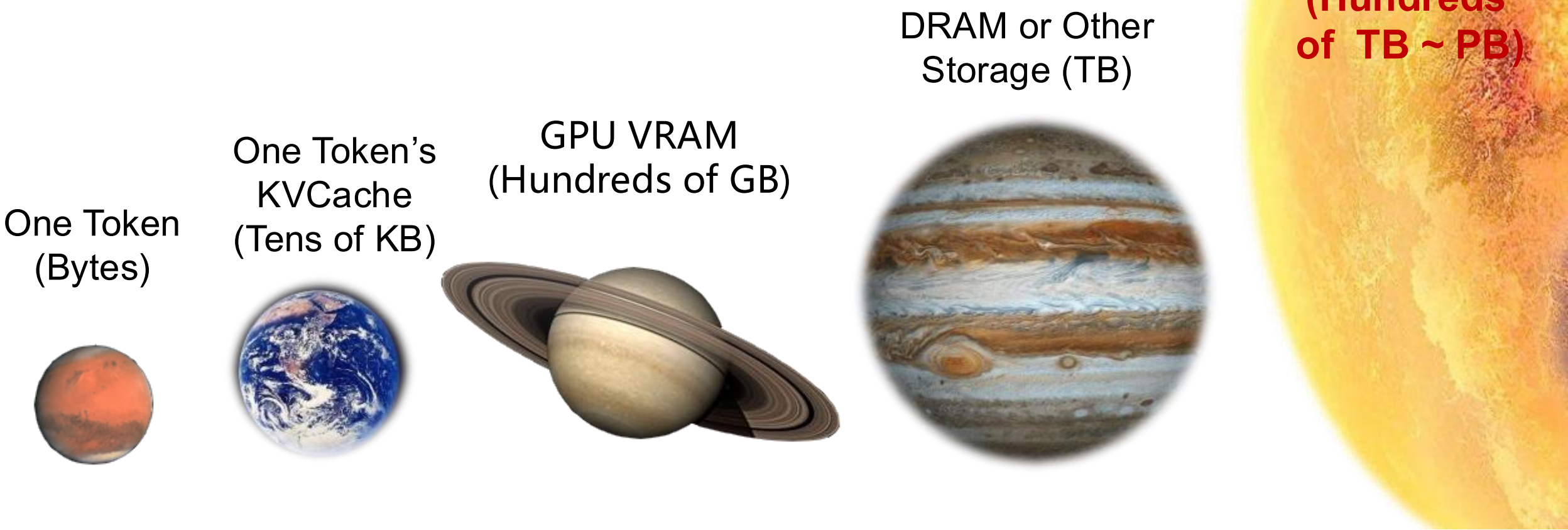
KVCache: Huge Challenge to Storage System

- The volume of reusable KVCache is much larger than the available storage capacity of a single inference node



KVCache: Huge Challenge to Storage System

- High transfer bandwidth required to avoid stall of GPU
  - 6 GB/s for A800 GPUs
  - 19 GB/s for H800 GPUs



Trading More Storage for Less Computation

Requirement 1

Large-capacity KVCache storage

Requirement 2

Low-latency and high-bandwidth KVCache transfer

Requirement 3

KVCache-aware scheduling

Trading More Storage for Less Computation

Requirement 1

High-capacity KVCache storage

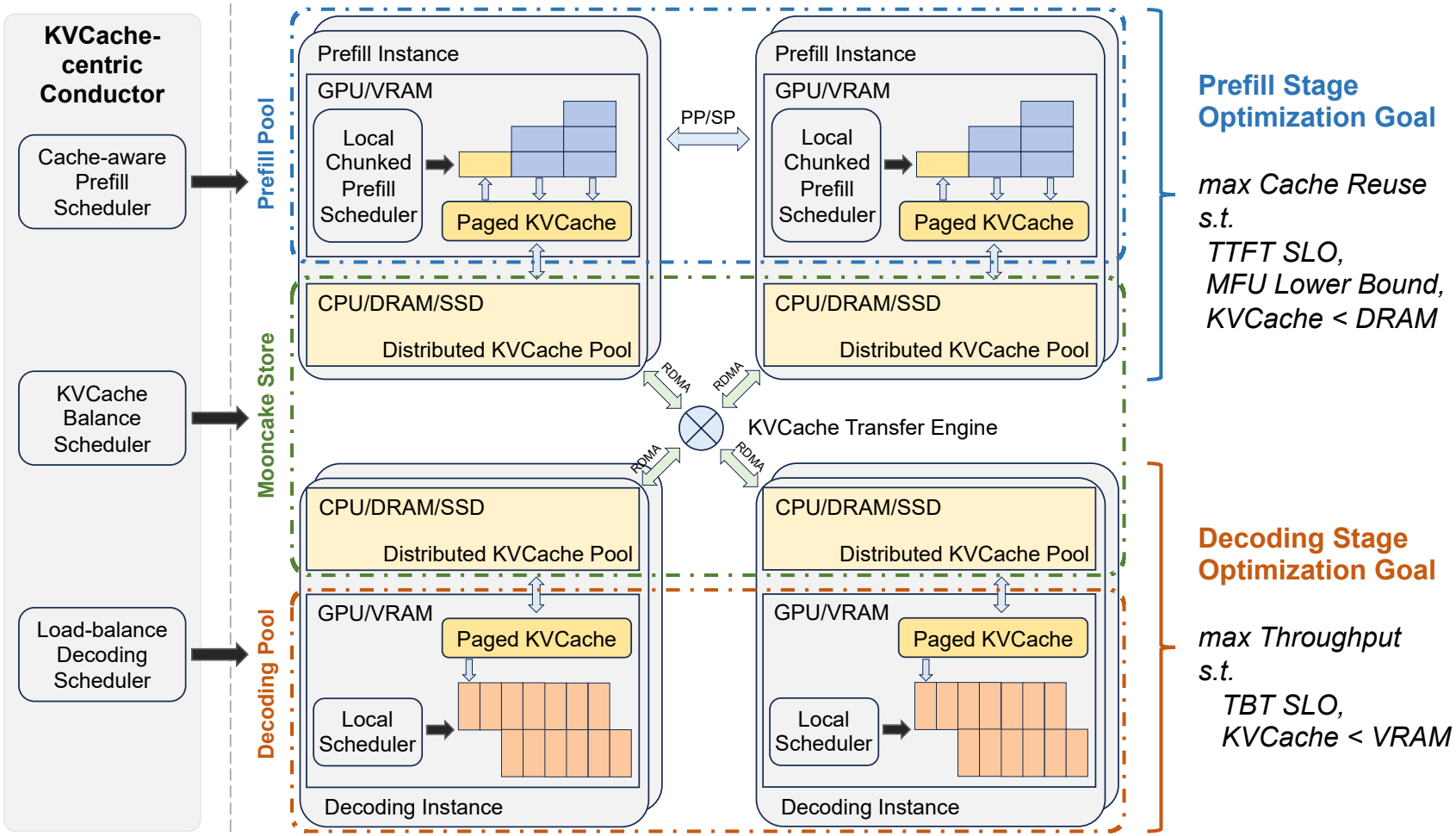
KVCache-centric Architecture is All You Need!

Requirement 3

KVCache-aware scheduling

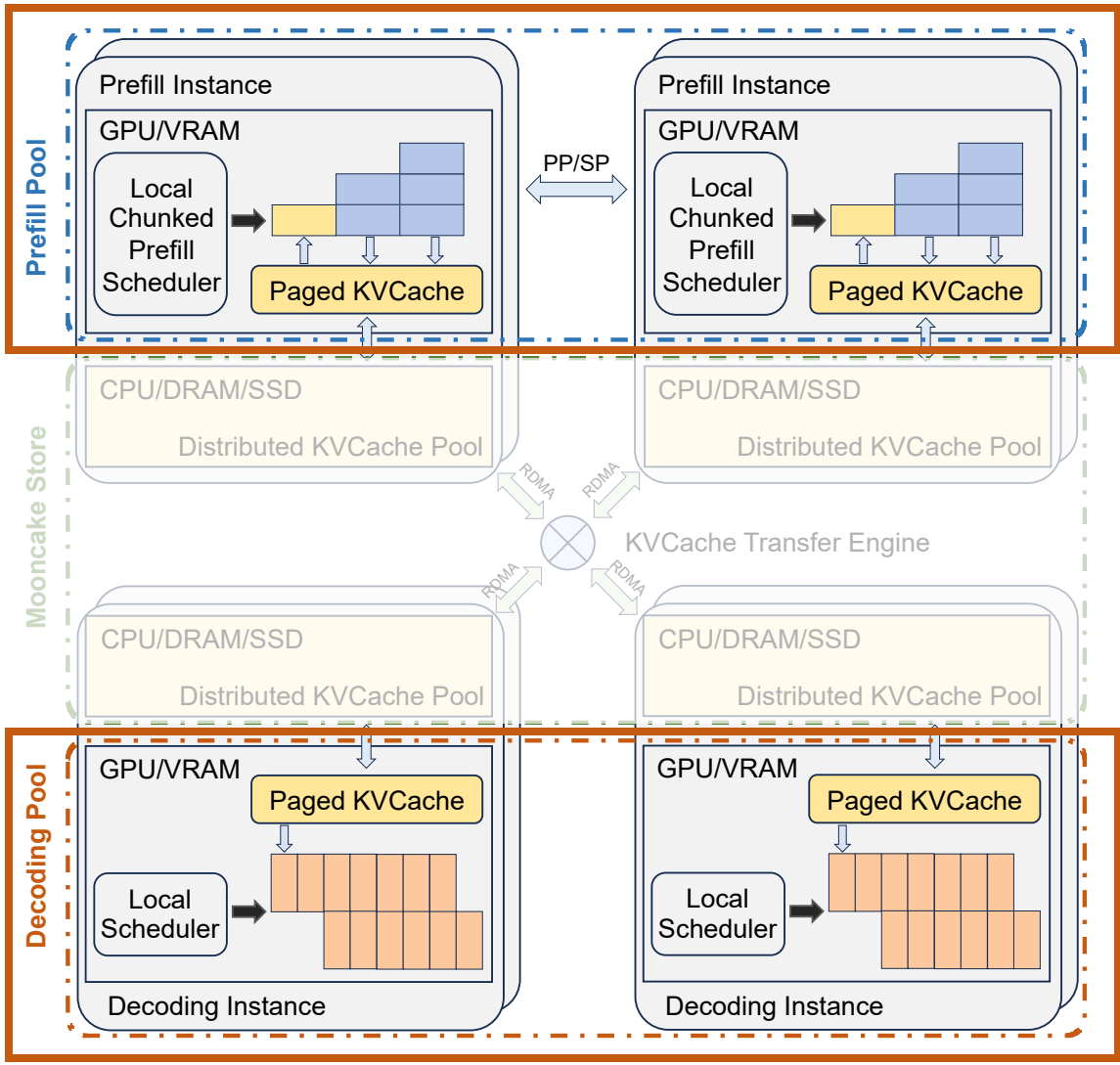
Mooncake: A KVCache-centric Disaggregated Architecture

- P&D disaggregation architecture centered around the distributed KVCache pool – Mooncake Store



Mooncake Store – Distributed Multi-layer KVCache Pool

- Large size and bandwidth (**Key of KVCache Storage**)
  - Distributed KVCache pool across all inference nodes
  - Utilize high performance connection like (GPUDirect) RDMA/Storage
  - Optimized for multi-NIC scenarios
- Design features
  - High scalability and fault tolerance
  - Independent to specific inference engine
  - Flexible API
- Performance
  - Achieve up to a 498% increase in effective request capacity
  - Save 29% - 61% on GPU computation costs



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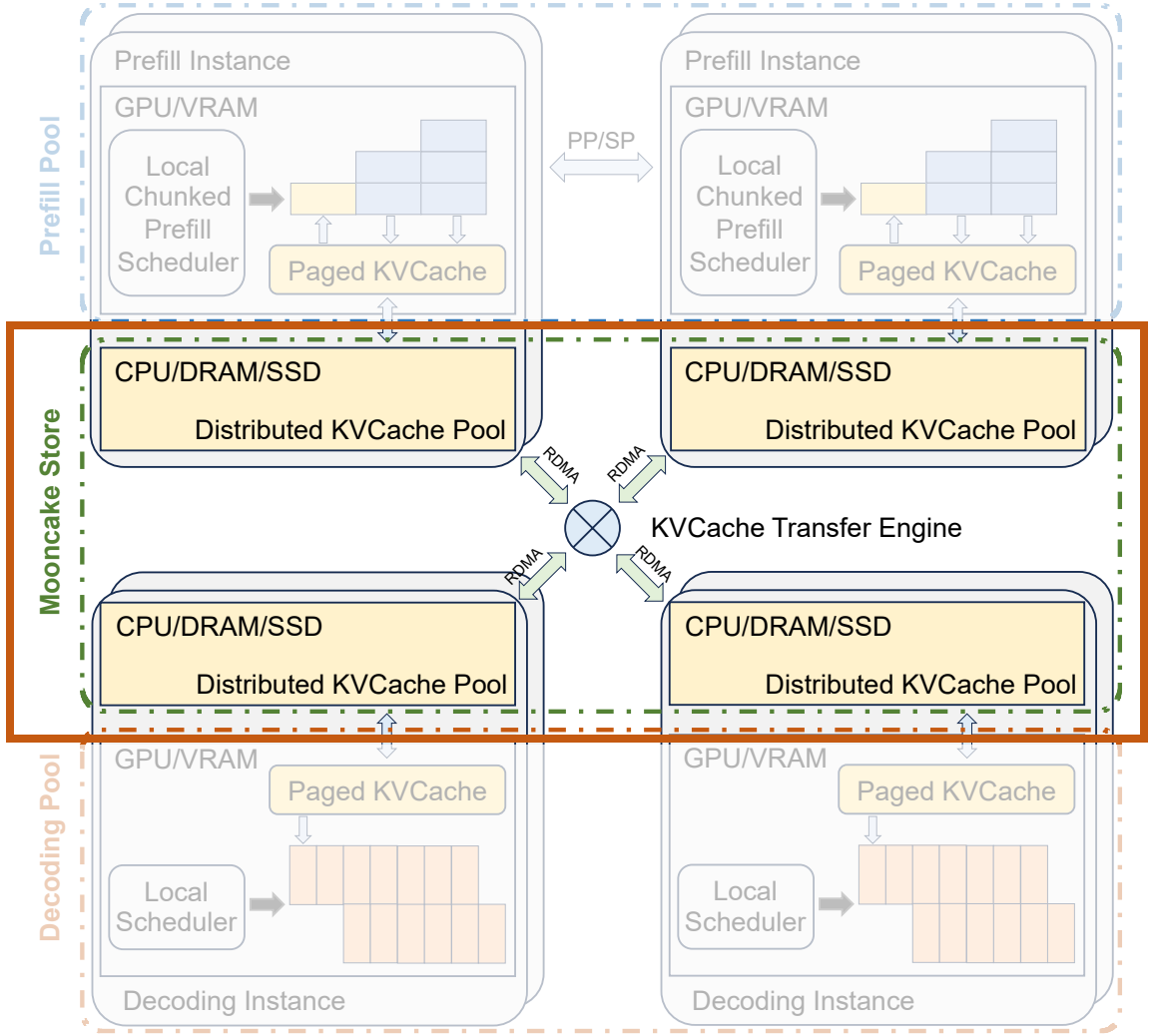
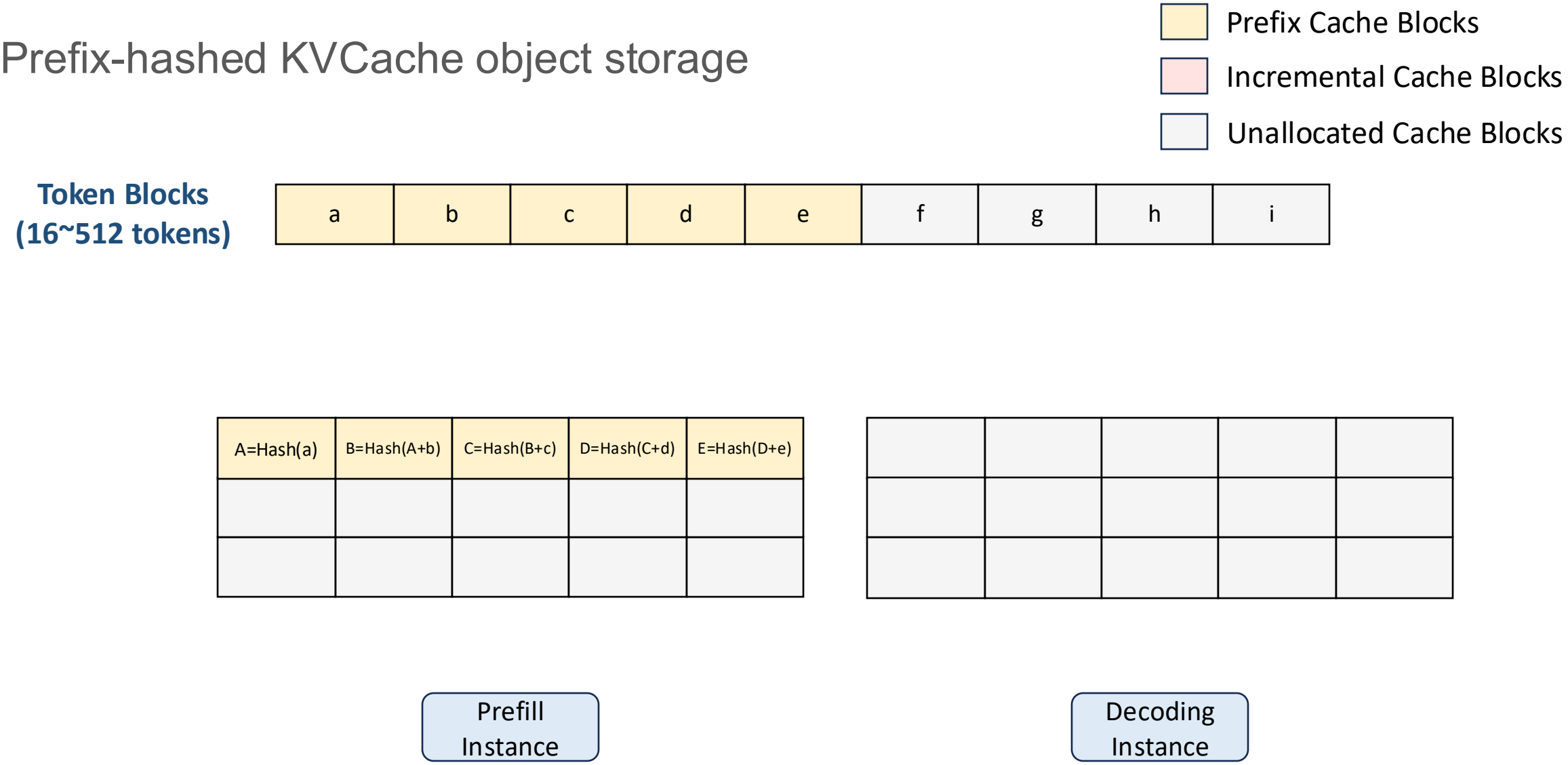


P&D Disaggregation and KVCache Management

- Prefix-hashed KVCache object storage



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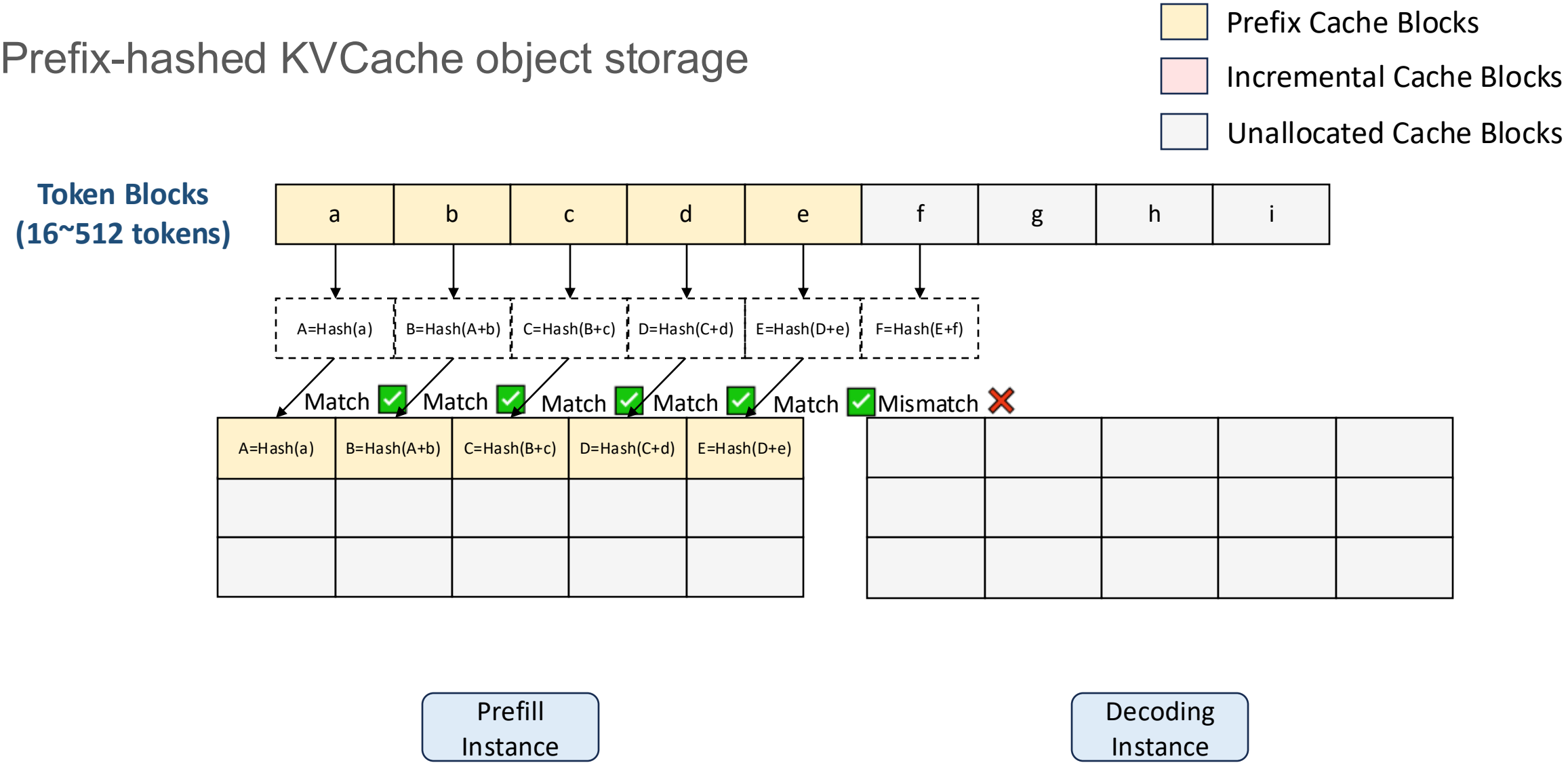

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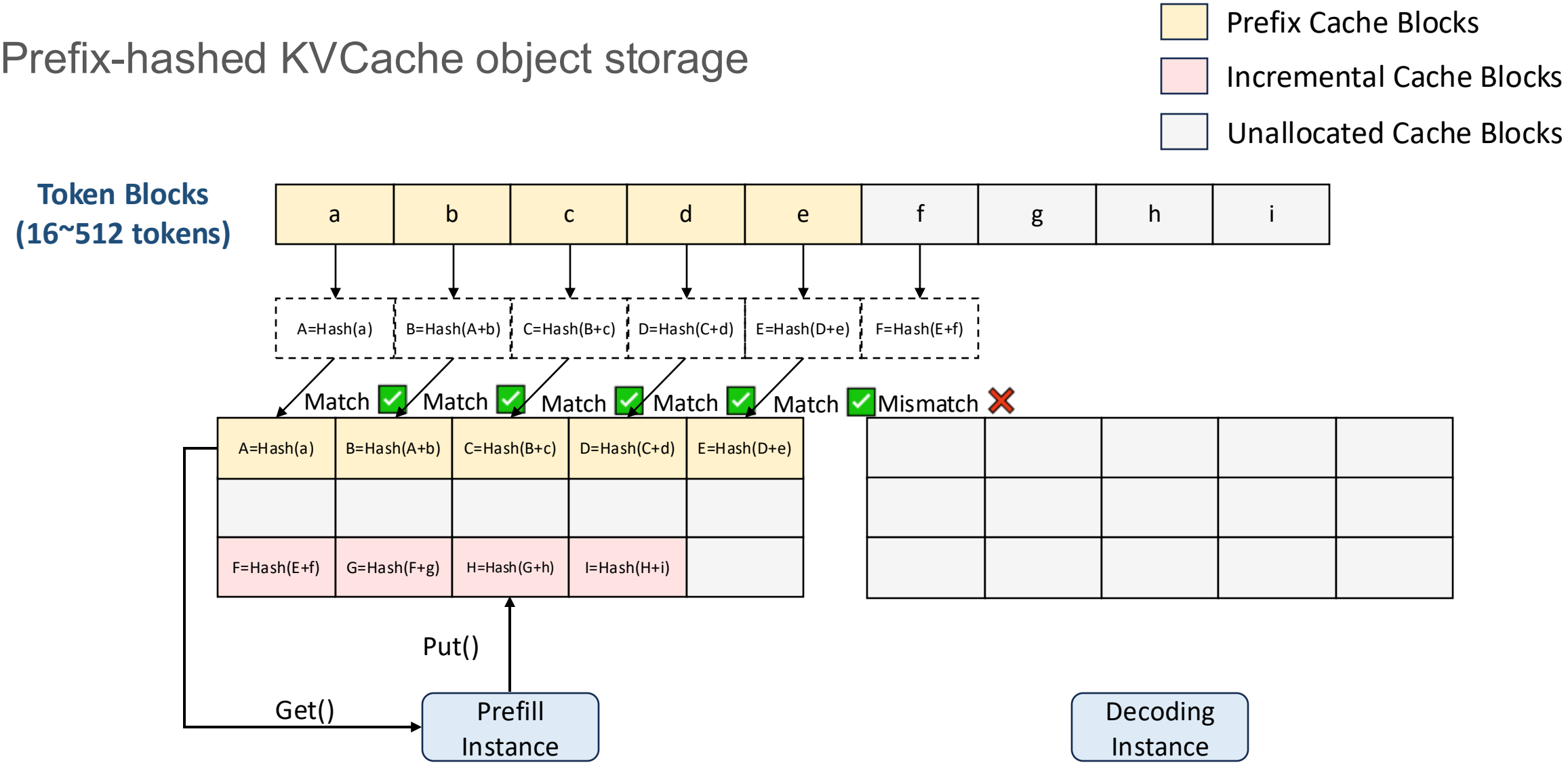

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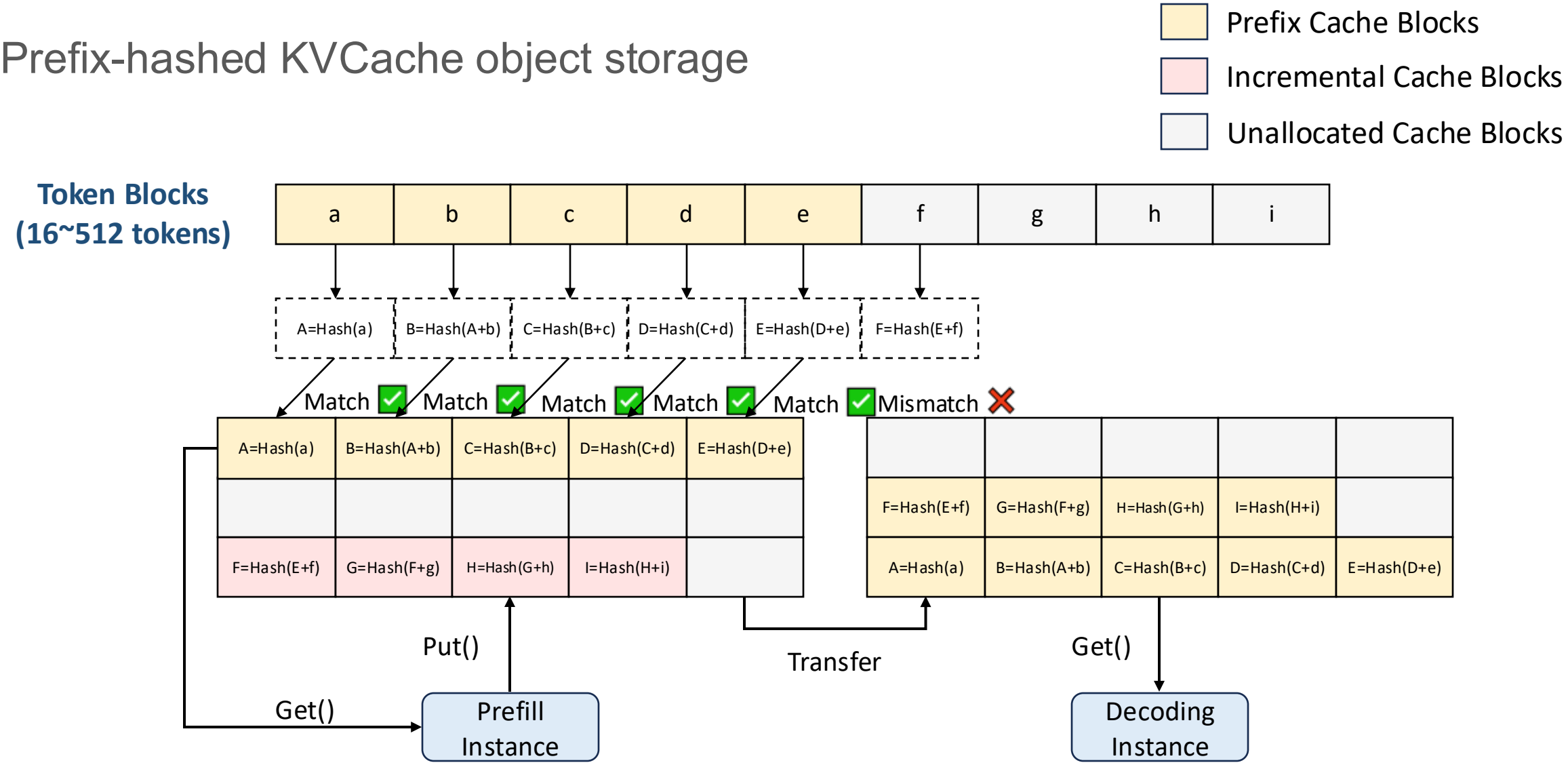
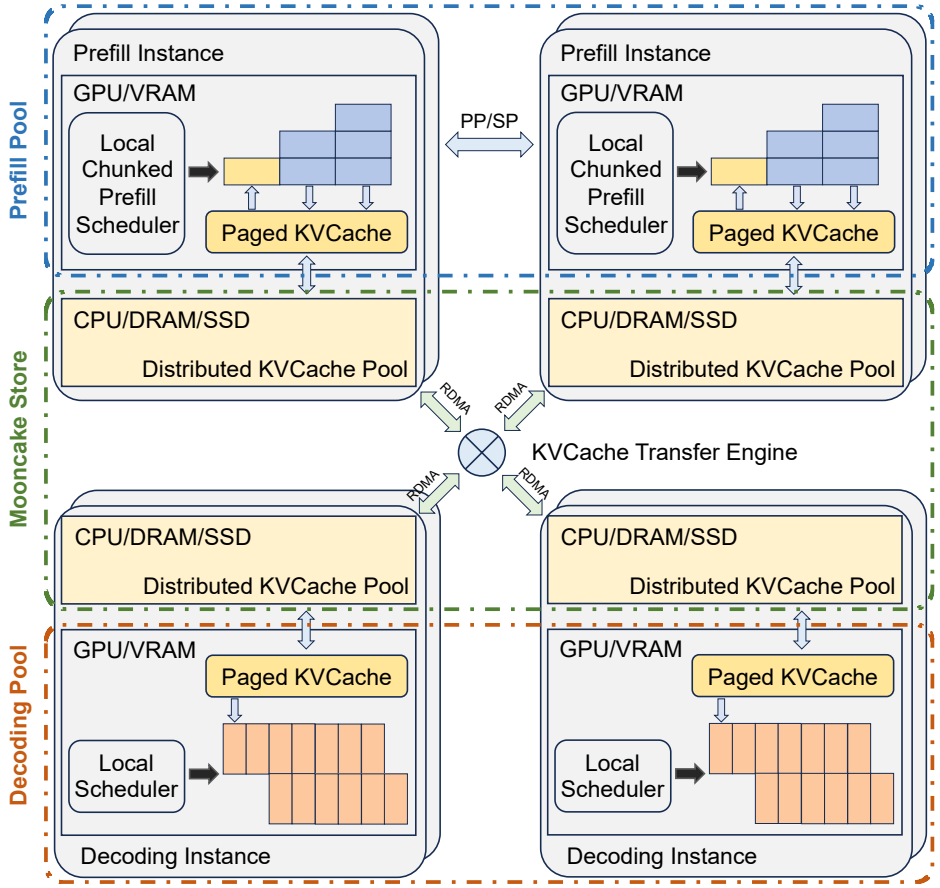
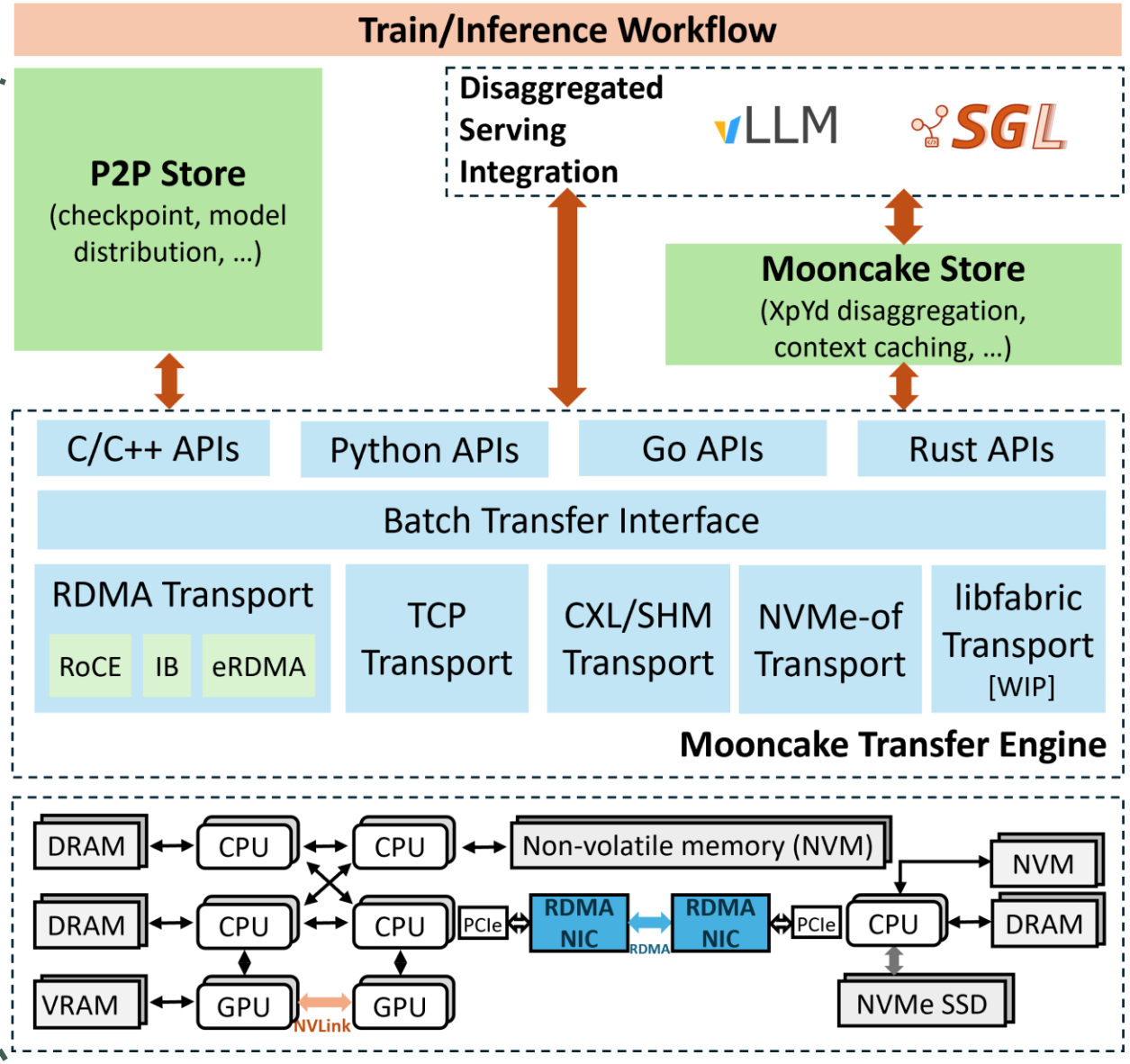


The Powerful Middleware of Distributed LLM Frameworks

- Mooncake Store is **open-source** and supports a variety of LLM scenarios, including PD disaggregation, KVCache reuse, and RL checkpoint distribution.





The Open Source Community

vLLM

- **vLLM** - The most widely used inference engine.
- Its distributed inference implementation is built on **Mooncake**.

v0.8.3 Latest

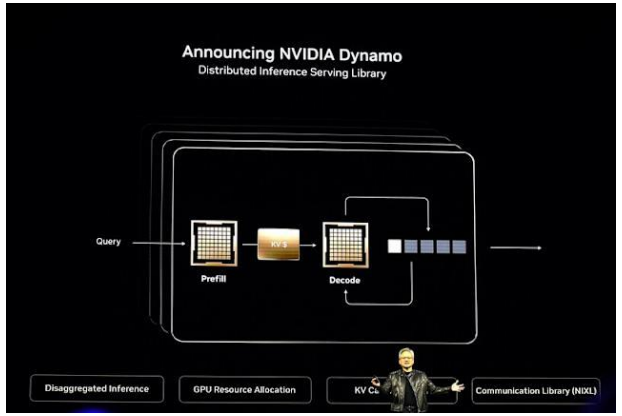
github-actions released this 2 days ago · 57 commits to main since this release v0.8.3 296c657

Cluster Scale Serving

- Support XpYd disaggregated prefill with MooncakeStore (#12957)

NVIDIA Dynamo

- **Dynamo** – the distributed inference system highlighted in the **GTC 2025 Keynote**.
- Its architecture is inspired by **Mooncake**, with explicit acknowledgment in its documentation.



Acknowledgement

We would like to acknowledge several open source software stacks for motivating us to create Dynamo.

- vLLM and vLLM-project
- SGLang
- DistServe
- Mooncake

- *Memory bottlenecks*: Large-scale inference workloads demand extensive capacity. KV cache offloading across memory hierarchies (HBM, DDR, N memory limits and speeds up latency. (Mooncake, AIBrix, LMCache)

SGL

- **SGLang** - the core inference engine of xAI, is widely used in DeepSeek's inference stack.
- Its distributed inference architecture is jointly developed in collaboration with **Mooncake**.

LMSYS Org @lmsysorg

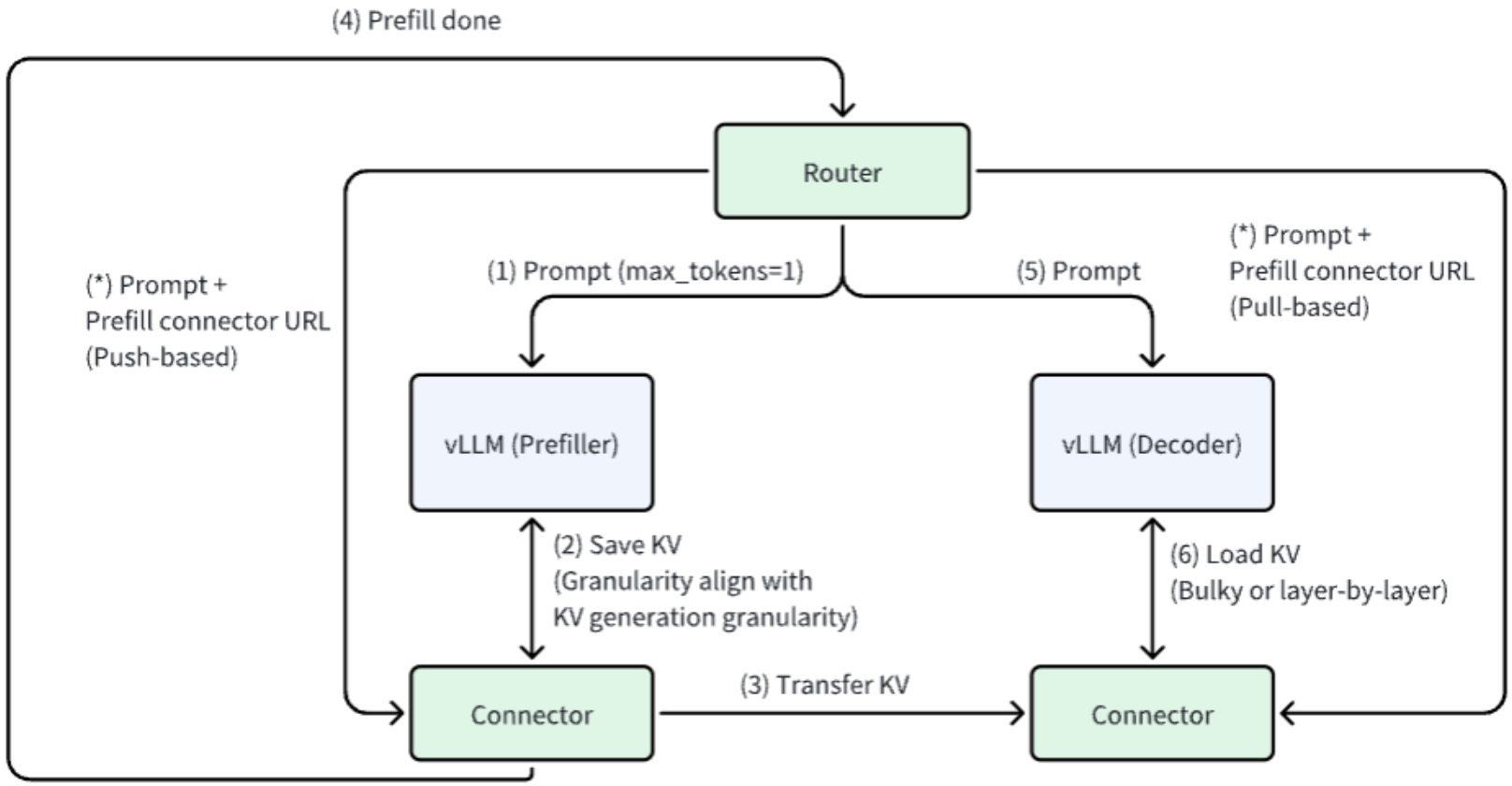
The SGLang Team is honored to announce that the following well-known companies and teams, among others, have adopted SGLang for running DeepSeek V3 and R1. @AMD @nvidia @Azure @basetenco @novita_ai_labs @BytedanceTalk @DataCrunch_io @hyperbolic_labs @Vultr @runPod

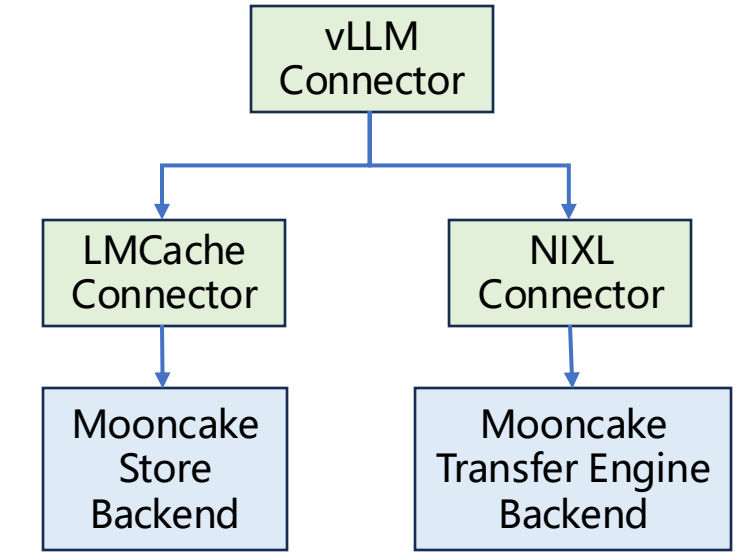
LMSYS Org @lmsysorg

SGLang has achieved a milestone with full support for PD Disaggregation, thanks to the MoonCake team. Teng Ma, Shangming Cai, Xuchun Shang and Yuan Luo were instrumental in this achievement. Special thanks to Atlas Cloud for their support with the H100s cluster. Let's go! 🚀

Mooncake + vLLM

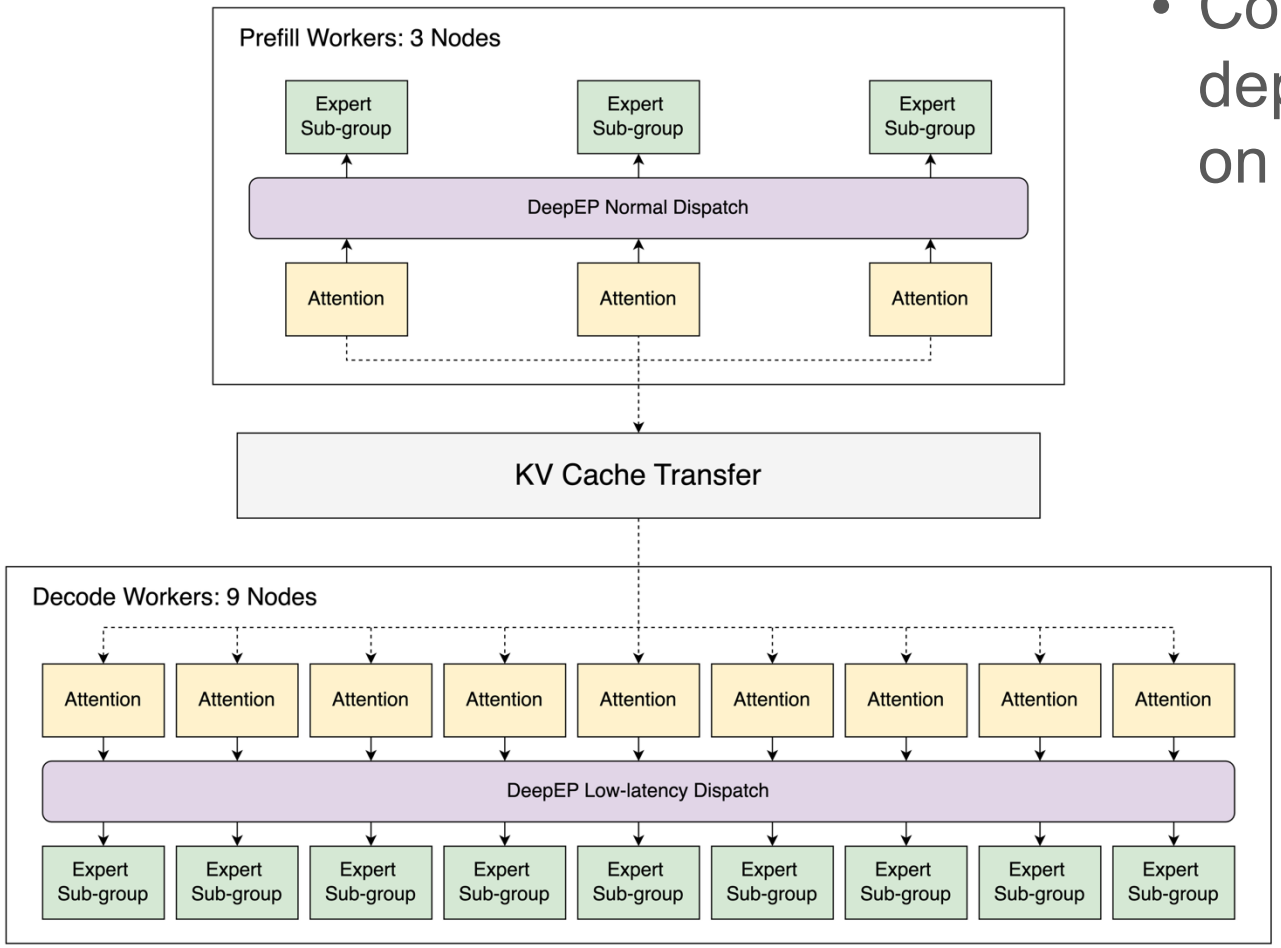
- Integrated with vLLM as the backend of the LMCache Connector[1]



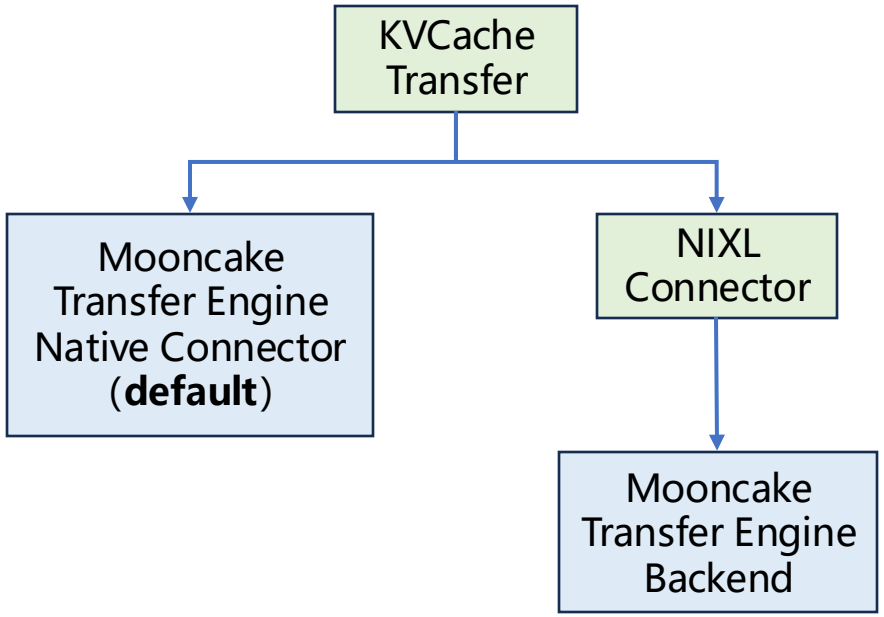


[1] https://blog.lmcache.ai/2025-05-08-mooncake

Mooncake + SGLang



- Collaborated with the SGLang team to deploy DeepSeek with PD-disaggregation on 96 H100 GPUs[1]



[1] https://lmsys.org/blog/2025-05-05-large-scale-ep

Mooncake + Dynamo

Add Mooncake backend in NIXL #169

Merged **tstamler** merged 19 commits into `ai-dynamo:main` from `alogfans:mooncake-support` 2 weeks ago

Conversation 26 | Commits 19 | Checks 5 | Files changed 13

alogfans commented on Apr 19 • edited — Contributor

Mooncake transfer engine is a high-performance, zero-copy data transfer library. To achieve better performance in NIXL, we have designed an new backend based on Mooncake transfer engine.

Usage

1. Build the install Mooncake manually:

```
git clone https://github.com/kvcache-ai/Mooncake.git
cd Mooncake
bash dependencies.sh
mkdir build
cd build
cmake .. -DBUILD_SHARED_LIBS=ON
make -j
sudo make install # compulsory
```

2. Build NIXL with option `enable_mooncake_backend` set as true
3. You can use Mooncake Backend if you create backend by `agent.createBackend("Mooncake", init_param, backend)`. See `examples/cpp/nixl_benchmark.cpp` as an example.

3 3

Add Mooncake backend in NIXL ... c7807b0

Join the Open Source Community for Efficient LLM Serving

kvcache.ai

KVCache.AI is a joint research project between MADSys and top industry collaborators, focusing on efficient LLM serving.

686 followers https://madsys.cs.tsinghua.edu.cn/ zhang_mingxing@mail.tsinghua.edu.cn

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Mooncake Public

Mooncake is the serving platform for Kimi, a leading LLM service provided by Moonshot AI.

C++ 3k 194

ktransformers Public

A Flexible Framework for Experiencing Cutting-edge LLM Inference Optimizations

Python 13.4k 925

https://github.com/kvcache-ai

Summary

Mooncake

Deployed System: Mooncake

- Trading more storage for less computation
- A KVCache-centric LLM serving architecture for higher throughput

Infrastructure: Mooncake Store

- Fast, scalable and fault-tolerant KVCache storage and transfer engine
- Open-sourced with support for multiple LLM serving frameworks

Benefits of Mooncake Store

- Efficient P&D disaggregation
- More KVCache reuse
- Flexible KVCache scheduling

Paper

GitHub